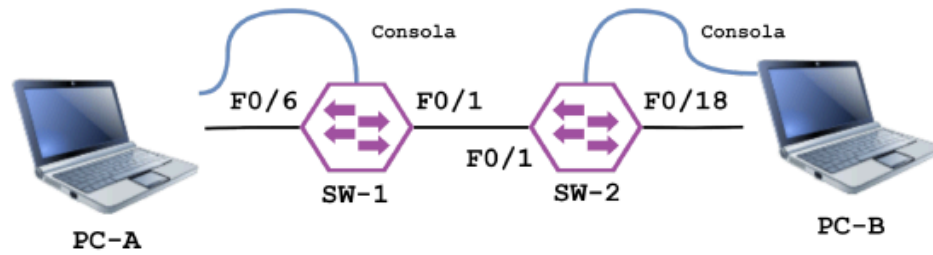


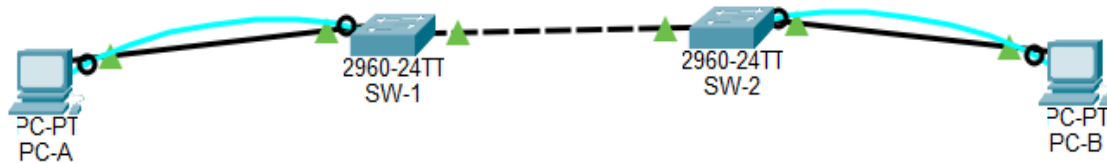
LAB 4 - Consigna 2



Con la siguiente Tabla de ruteo:

Device	Interface	IP Address	Subnet Mask	Default Gateway
SW-1	VLAN 1	192.168.1.11	255.255.255.0	N/A
SW-2	VLAN 1	192.168.1.12	255.255.255.0	N/A
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.10.4	255.255.255.0	192.168.10.1

Diseño en Cisco Packet Tracer



Desarrollo

- a) Desde cada computadora, ingresar a la terminal y configurar los switch. Nombrar a los mismos sw1 y sw2 respectivamente.

```
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname sw1
sw1(config)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname sw2
sw2(config)#
```

- b) Asignar contraseñas privilegiadas, de consola y vty.

```
sw1(config)#enable secret contrasena_exec
sw1(config)#line console 0
sw1(config-line)#password contrasena_consola
sw1(config-line)#login
sw1(config-line)#exit
sw1(config)#line vty 0 15
sw1(config-line)#password contrasena_vty
sw1(config-line)#login
sw1(config-line)#exit
sw1(config)#
```

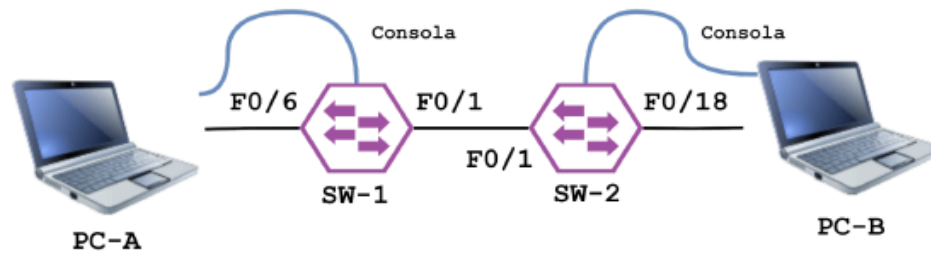
```
sw2(config)#enable secret contrasena_exec
sw2(config)#line console 0
sw2(config-line)#password contrasena_consola
sw2(config-line)#login
sw2(config-line)#exit
sw2(config)#line vty 0 15
sw2(config-line)#password contrasena_vty
sw2(config-line)#login
sw2(config-line)#exit
sw2(config)#
```

- c) Encriptar las contraseñas

```
sw1(config)#service password-encryption
```

```
sw2(config)#service password-encryption
```

- d) Configurar las redes VLAN para ambos switch según la tabla de direcciones provista.



Con la siguiente Tabla de ruteo:

Device	Interface	IP Address	Subnet Mask	Default Gateway
SW-1	VLAN 1	192.168.1.11	255.255.255.0	N/A
SW-2	VLAN 1	192.168.1.12	255.255.255.0	N/A
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.10.4	255.255.255.0	192.168.10.1

```
sw1(config)#interface vlan 1
sw1(config-if)#ip address 192.168.1.11 255.255.255.0
sw1(config-if)#no shutdown

sw1(config-if)#exit
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

```
sw2(config)#interface vlan 1
sw2(config-if)#ip address 192.168.1.12 255.255.255.0
sw2(config-if)#no shutdown

sw2(config-if)#exit
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

e) Desconectar todas las interfaces que no estén siendo utilizadas

1. *Veo que interfaces no están siendo utilizadas con “do show ip interface brief”:*

```
sw1(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/2 unassigned      YES manual down        down
FastEthernet0/3 unassigned      YES manual down        down
FastEthernet0/4 unassigned      YES manual down        down
FastEthernet0/5 unassigned      YES manual down        down
FastEthernet0/6 unassigned      YES manual up          up
FastEthernet0/7 unassigned      YES manual down        down
FastEthernet0/8 unassigned      YES manual down        down
FastEthernet0/9 unassigned      YES manual down        down
FastEthernet0/10 unassigned      YES manual down        down
FastEthernet0/11 unassigned      YES manual down        down
FastEthernet0/12 unassigned      YES manual down        down
FastEthernet0/13 unassigned      YES manual down        down
FastEthernet0/14 unassigned      YES manual down        down
FastEthernet0/15 unassigned      YES manual down        down
FastEthernet0/16 unassigned      YES manual down        down
FastEthernet0/17 unassigned      YES manual down        down
FastEthernet0/18 unassigned      YES manual down        down
FastEthernet0/19 unassigned      YES manual down        down
FastEthernet0/20 unassigned      YES manual down        down
FastEthernet0/21 unassigned      YES manual down        down
--More-- |
```

```
sw2(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/2 unassigned      YES manual down        down
FastEthernet0/3 unassigned      YES manual down        down
FastEthernet0/4 unassigned      YES manual down        down
FastEthernet0/5 unassigned      YES manual down        down
FastEthernet0/6 unassigned      YES manual down        down
FastEthernet0/7 unassigned      YES manual down        down
FastEthernet0/8 unassigned      YES manual down        down
FastEthernet0/9 unassigned      YES manual down        down
FastEthernet0/10 unassigned      YES manual down        down
FastEthernet0/11 unassigned      YES manual down        down
FastEthernet0/12 unassigned      YES manual down        down
FastEthernet0/13 unassigned      YES manual down        down
FastEthernet0/14 unassigned      YES manual down        down
FastEthernet0/15 unassigned      YES manual down        down
FastEthernet0/16 unassigned      YES manual down        down
FastEthernet0/17 unassigned      YES manual down        down
FastEthernet0/18 unassigned      YES manual up          up
FastEthernet0/19 unassigned      YES manual down        down
FastEthernet0/20 unassigned      YES manual down        down
FastEthernet0/21 unassigned      YES manual down        down
--More--
```


3. Ahora veo devuelta:

```
sw1(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual up up
FastEthernet0/2 unassigned YES manual administratively down down
FastEthernet0/3 unassigned YES manual administratively down down
FastEthernet0/4 unassigned YES manual administratively down down
FastEthernet0/5 unassigned YES manual administratively down down
FastEthernet0/6 unassigned YES manual up up
FastEthernet0/7 unassigned YES manual administratively down down
FastEthernet0/8 unassigned YES manual administratively down down
FastEthernet0/9 unassigned YES manual administratively down down
FastEthernet0/10 unassigned YES manual administratively down down
FastEthernet0/11 unassigned YES manual administratively down down
FastEthernet0/12 unassigned YES manual administratively down down
FastEthernet0/13 unassigned YES manual administratively down down
FastEthernet0/14 unassigned YES manual administratively down down
FastEthernet0/15 unassigned YES manual administratively down down
FastEthernet0/16 unassigned YES manual administratively down down
FastEthernet0/17 unassigned YES manual administratively down down
FastEthernet0/18 unassigned YES manual administratively down down
FastEthernet0/19 unassigned YES manual administratively down down
FastEthernet0/20 unassigned YES manual administratively down down
FastEthernet0/21 unassigned YES manual administratively down down
FastEthernet0/22 unassigned YES manual administratively down down
FastEthernet0/23 unassigned YES manual administratively down down
FastEthernet0/24 unassigned YES manual administratively down down
GigabitEthernet0/1 unassigned YES manual administratively down down
GigabitEthernet0/2 unassigned YES manual administratively down down
Vlan1 192.168.1.11 YES manual up up
sw1(config)#
```

```
sw2(config-if-range)#exit
sw2(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual up up
FastEthernet0/2 unassigned YES manual administratively down down
FastEthernet0/3 unassigned YES manual administratively down down
FastEthernet0/4 unassigned YES manual administratively down down
FastEthernet0/5 unassigned YES manual administratively down down
FastEthernet0/6 unassigned YES manual administratively down down
FastEthernet0/7 unassigned YES manual administratively down down
FastEthernet0/8 unassigned YES manual administratively down down
FastEthernet0/9 unassigned YES manual administratively down down
FastEthernet0/10 unassigned YES manual administratively down down
FastEthernet0/11 unassigned YES manual administratively down down
FastEthernet0/12 unassigned YES manual administratively down down
FastEthernet0/13 unassigned YES manual administratively down down
FastEthernet0/14 unassigned YES manual administratively down down
FastEthernet0/15 unassigned YES manual administratively down down
FastEthernet0/16 unassigned YES manual administratively down down
FastEthernet0/17 unassigned YES manual administratively down down
FastEthernet0/18 unassigned YES manual up up
FastEthernet0/19 unassigned YES manual administratively down down
FastEthernet0/20 unassigned YES manual administratively down down
FastEthernet0/21 unassigned YES manual administratively down down
FastEthernet0/22 unassigned YES manual administratively down down
FastEthernet0/23 unassigned YES manual administratively down down
FastEthernet0/24 unassigned YES manual administratively down down
GigabitEthernet0/1 unassigned YES manual administratively down down
GigabitEthernet0/2 unassigned YES manual administratively down down
Vlan1 192.168.1.12 YES manual up up
sw2(config)#
```

f) Guardar la configuración (write memory)

```
sw1(config)#end
sw1#
%SYS-5-CONFIG_I: Configured from console by console

sw1#write memory
Building configuration...
[OK]
sw1#
```

```
sw2(config)#end
sw2#
%SYS-5-CONFIG_I: Configured from console by console

sw2#write memory
Building configuration...
[OK]
sw2#
```

g) Testear comunicación usando pings entre las computadoras.

```
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

RESULTADO: Exitoso

- Debido a que el estado inicial de las interfaces están por defecto en VLAN 1 (default)

h) Crear VLANs en ambos switches.

```
sw1(config)#vlan 10
sw1(config-vlan)#name Laboratorio
sw1(config-vlan)#vlan 20
sw1(config-vlan)#name Bar
sw1(config-vlan)#vlan 99
sw1(config-vlan)#name Management
sw1(config-vlan)#end
sw1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
sw2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sw2(config)#vlan 10
sw2(config-vlan)#name Laboratorio
sw2(config-vlan)#vlan 20
sw2(config-vlan)#name Bar
sw2(config-vlan)#vlan 99
sw2(config-vlan)#name Management
sw2(config-vlan)#end
```

i) Utilizar show vlan brief para visualizar la lista de VLANs en alguno de los switch.
¿Cuál es la VLAN utilizada por defecto?

```
sw1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Laboratorio	active	
20	Bar	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
sw2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Laboratorio	active	
20	Bar	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- La VLAN utilizada por defecto es la **VLAN 1** (llamada **default**)

- j) Asignar la PC-A a la VLAN Laboratorio.

```
sw1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sw1(config)#interface f0/6
sw1(config-if)#switchport mode access
sw1(config-if)#switchport access vlan 10
```

- k) Desde la VLAN 1, remover la ip de Management y configurarla para funcionar en la VLAN 99 (que configuramos como Management).

```
sw1(config-if)#exit
sw1(config)#interface vlan 1
sw1(config-if)#no ip address
sw1(config-if)#interface vlan 99
sw1(config-if)#ip address 192.168.1.11 255.255.255.0
sw1(config-if)#no shutdown
sw1(config-if)#end
%LINK-5-CHANGED: Interface Vlan99, changed state to up
```

- l) Verificar el estado de la VLAN utilizando show vlan brief y el estado de las interfaces utilizando show ip interface brief.

1. Veo las VLANs:

```
sw1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Laboratorio	active	Fa0/6
20	Bar	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- El puerto Fa0/6 ahora aparece asociado a la VLAN 10 (Laboratorio).

2. Veo las interfaces:

```
sw1#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/1          unassigned      YES manual up             up
FastEthernet0/2          unassigned      YES manual administratively down down
FastEthernet0/3          unassigned      YES manual administratively down down
FastEthernet0/4          unassigned      YES manual administratively down down
FastEthernet0/5          unassigned      YES manual administratively down down
FastEthernet0/6          unassigned      YES manual up             up
FastEthernet0/7          unassigned      YES manual administratively down down
FastEthernet0/8          unassigned      YES manual administratively down down
FastEthernet0/9          unassigned      YES manual administratively down down
FastEthernet0/10         unassigned      YES manual administratively down down
FastEthernet0/11         unassigned      YES manual administratively down down
FastEthernet0/12         unassigned      YES manual administratively down down
FastEthernet0/13         unassigned      YES manual administratively down down
FastEthernet0/14         unassigned      YES manual administratively down down
FastEthernet0/15         unassigned      YES manual administratively down down
FastEthernet0/16         unassigned      YES manual administratively down down
FastEthernet0/17         unassigned      YES manual administratively down down
FastEthernet0/18         unassigned      YES manual administratively down down
FastEthernet0/19         unassigned      YES manual administratively down down
FastEthernet0/20         unassigned      YES manual administratively down down
FastEthernet0/21         unassigned      YES manual administratively down down
FastEthernet0/22         unassigned      YES manual administratively down down
FastEthernet0/23         unassigned      YES manual administratively down down
FastEthernet0/24         unassigned      YES manual administratively down down
GigabitEthernet0/1       unassigned      YES manual administratively down down
GigabitEthernet0/2       unassigned      YES manual administratively down down
Vlan1                    unassigned      YES manual up             up
Vlan99                   192.168.1.11   YES manual up             down
sw1#
```

- La VLAN 99 figura con protocolo down. Esto sucede porque una interfaz virtual necesita al menos un puerto físico activo asignado a esa VLAN o un enlace troncal activo transportándola para ponerse en up.

m) Asignar la PC-B a la VLAN Laboratorio en el sw2. Repetir el inciso k) pero para el sw2.

```
sw2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sw2(config)#interface f0/18
sw2(config-if)#switchport mode access
sw2(config-if)#switchport access vlan 10
sw2(config-if)#exit
sw2(config)#interface vlan 1
sw2(config-if)#no ip address
sw2(config-if)#interface vlan 99
sw2(config-if)#ip address 192.168.1.12 255.255.255.0
sw2(config-if)#no shutdown
sw2(config-if)#end
%LINK-5-CHANGED: Interface Vlan99, changed state to up
```

```
sw2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Laboratorio	active	Fa0/18
20	Bar	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
sw2(config)#do show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	administratively down	down
FastEthernet0/3	unassigned	YES	manual	administratively down	down
FastEthernet0/4	unassigned	YES	manual	administratively down	down
FastEthernet0/5	unassigned	YES	manual	administratively down	down
FastEthernet0/6	unassigned	YES	manual	administratively down	down
FastEthernet0/7	unassigned	YES	manual	administratively down	down
FastEthernet0/8	unassigned	YES	manual	administratively down	down
FastEthernet0/9	unassigned	YES	manual	administratively down	down
FastEthernet0/10	unassigned	YES	manual	administratively down	down
FastEthernet0/11	unassigned	YES	manual	administratively down	down
FastEthernet0/12	unassigned	YES	manual	administratively down	down
FastEthernet0/13	unassigned	YES	manual	administratively down	down
FastEthernet0/14	unassigned	YES	manual	administratively down	down
FastEthernet0/15	unassigned	YES	manual	administratively down	down
FastEthernet0/16	unassigned	YES	manual	administratively down	down
FastEthernet0/17	unassigned	YES	manual	administratively down	down
FastEthernet0/18	unassigned	YES	manual	up	up
FastEthernet0/19	unassigned	YES	manual	administratively down	down
FastEthernet0/20	unassigned	YES	manual	administratively down	down
FastEthernet0/21	unassigned	YES	manual	administratively down	down
FastEthernet0/22	unassigned	YES	manual	administratively down	down
FastEthernet0/23	unassigned	YES	manual	administratively down	down
FastEthernet0/24	unassigned	YES	manual	administratively down	down
GigabitEthernet0/1	unassigned	YES	manual	administratively down	down
GigabitEthernet0/2	unassigned	YES	manual	administratively down	down
Vlan1	unassigned	YES	manual	up	up
Vlan99	192.168.1.12	YES	manual	up	down

- n) Verificar la conectividad entre PC-A y PC-B utilizando pings. Verificar conectividad entre sw1 y sw2 utilizando pings. Interpretar los resultados.

1. Hago la prueba de conectividad entre ambas pc y switches utilizando pings:

(a) *Prueba entre PCs:*

```
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

(b) *Prueba entre Switches:*

```
sw1#ping 192.168.1.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.12, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

RESULTADO: Ambas fallidas

- Aunque ambas PCs están asignadas lógicamente a la misma VLAN 10 (Laboratorio), el enlace que conecta sw1 con sw2 (Fa0/1) sigue operando como un enlace de acceso común en la VLAN 1 y no como un enlace troncal 802.1Q en modo trunk.

2. Para solucionarlo, configuramos el puerto Fa0/1 en modo **trunk**, como troncal IEEE 802.1Q, para posibilitar el tráfico multi-VLAN:

```
sw1(config)#interface f0/1
sw1(config-if)#switchport mode trunk

sw1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

3. Ahora, haciendo la misma prueba de conectividad entre ambas PC y SWs con pings:

```
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
sw1#ping 192.168.1.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.12, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms
```

RESULTADO: Exitoso

- Los pings responden con 100% de efectividad, existe conectividad entre PC-A y PC-B y entre SW-1 y SW-2.